

Field observation and cause investigation of low-frequency cable vibrations in a cable-stayed bridge

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ABSTRACT

Considerable levels of stay cable vibration in cable-supported bridges pose significant challenges for safety, operation, and maintenance. An accurate understanding of the specific causes and types of vibration is essential for effectively addressing this problem with an optimal design for mitigation measures. This study aims to investigate the cause of an extraordinary low-frequency vibration that occurred at a stay cable by employing comprehensive methodologies. The modal properties and vibrational characteristics of cable vibrations were first obtained through finite element modeling, operational modal analysis, and vision-based displacement measurements. Subsequently, the primary cause of low-frequency vibrations was investigated using field observations and wind tunnel tests. Field-monitoring data revealed that stay cables were excited by small deck vibrations. The critical wind speeds at which deck vibrations occurred in a wind tunnel experiment were aligned with the ranges of wind speeds that produce low-frequency cable vibrations, which confirmed that these two phenomena are manifested under similar wind conditions. A calculated support excitation amplitude, determined via the use of an empirical equation, exhibited consistency with vision-based measured displacement obtained in field experiments. These findings underscore the potential risk of support excitation between minor deck vibrations and cables, which provides valuable insights that could mitigate cable vibrations.

1. Introduction

The excessive vibration of cables presents a pivotal challenge to the operational and maintenance aspects of cable-supported bridges. Vibration results mainly from wind — rain-wind-induced vibrations, vortex-induced vibrations (VIVs), and diverse galloping occurrences — and these are predominantly driven by the high flexibility and low-damping capacity of cables [11,27,6,7,8]. Effectively mitigating cable

vibrations inherently requires precise identification of the specific causes and types, because the intricate characteristics of cable vibrations have significant implications for the optimal design of mitigation devices. In this regard, recent research has increasingly applied field monitoring and experimental methodologies in order to understand the causes of such vibrational phenomena. For example, Zuo and Jones [28] investigated wind- and rain-wind-induced vibrations in cable-stayed bridges using long-term measurements. Additionally, Ni et al. (2007)

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examined large-amplitude cable oscillations in the Dongting Lake Bridge under rain-wind conditions, and they elucidated factors such as critical wind velocity and yaw angle ranges based on field observations in light-to-moderate rain. For the Kuimen Bridge, Miao et al. [25] explored the large-amplitude vibrations in stay cables induced by the installation of rectangular lamps by highlighting the rectangular shape of aluminum alloy boxes of the lamps and changing the sectional shape to mitigate the instability caused by galloping vibrations at low wind speeds. Zhang et al. [27] identified the causes of exceptional wind-induced vibrations in ultra-long stay cables by using monitoring data that included mean wind velocity, dynamic cable responses, and modal parameters. These studies exhibit the efficacy of structural health monitoring (SHM) systems in assessing harmful vibrations under service conditions.

In addition to wind-induced vibrations, cables in cable-stayed bridges can experience support excitation and parametric excitation. Support excitation arises from the direct movements of the bridge supports [2,24]. When the frequency of support movements aligns with the natural frequency of the cables, large amplitude of oscillations can occur. These movements can be analyzed using resonance theory. This type of excitation can cause both in-plane and out-of-plane vibrations, depending on the direction of the support movements. Recent studies have shown that increasing support stiffness and optimizing damper performance can effectively control cable vibrations caused by support excitation [12,4].

Conversely, parametric excitation is caused by small periodic movements of the deck or pylons that induce axial motion in the cable supports [15,23,3,5]. This form of excitation is inherently non-linear, stemming from time-varying system parameters such as variations in cable tension due to periodic deck movements. Parametric excitation occurs when the excitation frequency is a multiple of the cable's natural frequency.

In 2019, despite implementing a wind-resistance design that included section model tests targeting vortex-induced vibrations (VIVs) and flutter instability, unexpected VIVs were frequently observed on the deck of a cable-stayed bridge. In addition to the bridge deck vibrations, the design phase also addressed cable vibrations, including rain-wind-induced vibration, dry galloping, wake galloping, VIV, parametric excitation, and support excitation using analytical approaches. A subsequent in-depth field investigation revealed that the improper installation of inspection rails under the steel deck was the main cause of the VIV. To address this issue, multiple tuned mass dampers (MTMD) were installed within the center of the main span, which successfully suppressed the first-vertical (V1) mode of VIV. However, additional reports emerged of noteworthy vibrations in several stay cables, and these were categorized into two types. The first type involves typical high-mode VIVs with frequencies approximating 20 Hz [16], which are primarily observed in most of the longer cables. Contrary to this, the second type produces vibrations of a lower frequency at less than 0.5 Hz in specific stay cables [22]. The observation of such low-frequency dominance is unconventional in the study of cable vibration, and a comprehensive investigation into the causes of this vibration was necessary before an effective mitigation strategy could be formulated.

The present study illustrates how the authors identified the vibration characteristics and pinpointed the primary causes of observed low-frequency cable vibrations. The modal properties of the bridge were first estimated using a well-developed finite element (FE) model, operational modal analysis (OMA), and vision-based vibration monitoring. The subsequent investigation into the main cause of low-frequency cable vibration involved field observations and wind tunnel tests. Specifically, wind conditions triggering deck vibration in a wind tunnel experiment were compared with those observed during low-frequency cable vibration to define their relationship. The vibrational amplitudes of excited cables were empirically determined to assess the potential and risks associated with cable vibration. This study's novelty lies in: (1) reporting a unique low-frequency cable vibration phenomenon observed in a cable-stayed bridge; (2) proposing a cause-investigation procedure that

combines data analysis from the built-in monitoring system with necessary wind tunnel tests and field tests, which successfully demonstrated that the phenomenon was rooted in the undetectable amount of VIV locked to the second mode of the bridge deck; and 3) offering managerial recommendations based on the investigation into the causes.

2. Investigated bridge

2.1. Bridge description

The cable-stayed bridge examined for this study was opened to traffic in 2019 with a main-span length of 510 m, as shown in Fig. 1(a). The center span is composed of a streamlined steel box deck with a width of 16.1 m that provides two lanes of traffic, as illustrated in Fig. 1(b). The side spans are composed of concrete box decks. The bridge adopts two different-height pylons with bundle-type tie-down stays in the approaching spans.

Fig. 1(a) depicts two stay cables, which are the main focus of discussion in this study. "Cable 36" is the longest cable and "Cable 34" is the second next cable to the longest. "Cable 34" was subjected to low-frequency vibration. The stay cables are symmetrically arranged on both sides of the deck cross-section, which results in a total of 108 strands. Specifically, the cables display an asymmetrical alignment along the bridge, where the stay cables at the side spans are bundled to the piers providing a unique appearance and structural efficiency. Gensui rubber dampers were employed for the relatively longer cables, while high-damping rubber (HDR) dampers were installed for the other cables to enhance the damping ratios. Furthermore, helical fillets were added to each cable to reduce rain-wind-induced vibrations.

The bridge was equipped with a built-in SHM system whereby an anemometer (Windmaster Pro, UK) was mounted 5 m above the deck surface to measure wind speed, and a pair of vertical accelerometers were placed on each side of the deck at the mid-point of the center span with a sampling frequency of 100 Hz. The acquired data from these sensors were stored in mass storage devices that maintained time synchronization. Additionally, closed-circuit television (CCTV) cameras positioned on the top of the pylon provided sequential images of the deck and cables and enabled engineers to determine the instances of cable vibration in a low-frequency mode. Fig. 2 features images from the CCTV with the target cables indicated.

2.2. Operational modal analysis

In order to estimate the modal parameters, the Natural Excitation Technique combined with an Eigensystem Realization Algorithm (NExT-ERA) was employed. A total of 28 h of accelerometer monitoring data-sets, measuring the deck's movements, were compiled from March 6, 2019, at 06:00 to March 7, 2019, at 10:00 to target the V1 and second vertical (V2) modes of the deck. NExT-ERA is a well-established output-only system identification algorithm known for its applicability to stationary white noise. The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm was herein used to group the modal estimates that were obtained as the system orders increased. While the application of DBSCAN offers the advantage of automating modal analysis, it inherently requires an overestimation of the system order, which could lead to the presence of spurious poles even after the removal of outliers during the DBSCAN process. To address this challenge, the authors additionally applied the finite element method-based finite impulse response filter to reconstruct the dynamic displacement from acceleration responses [18,9]. By considering the physical relationship between displacement and acceleration, this reconstruction naturally filters out high-frequency components, allowing the fundamental mode to dominate. This enhancement helps mitigate the issue of spurious poles. Details of the algorithms for the improved NExT-ERA are provided in the relevant papers [10].

Fig. 3 illustrates the distribution of modal frequencies and damping

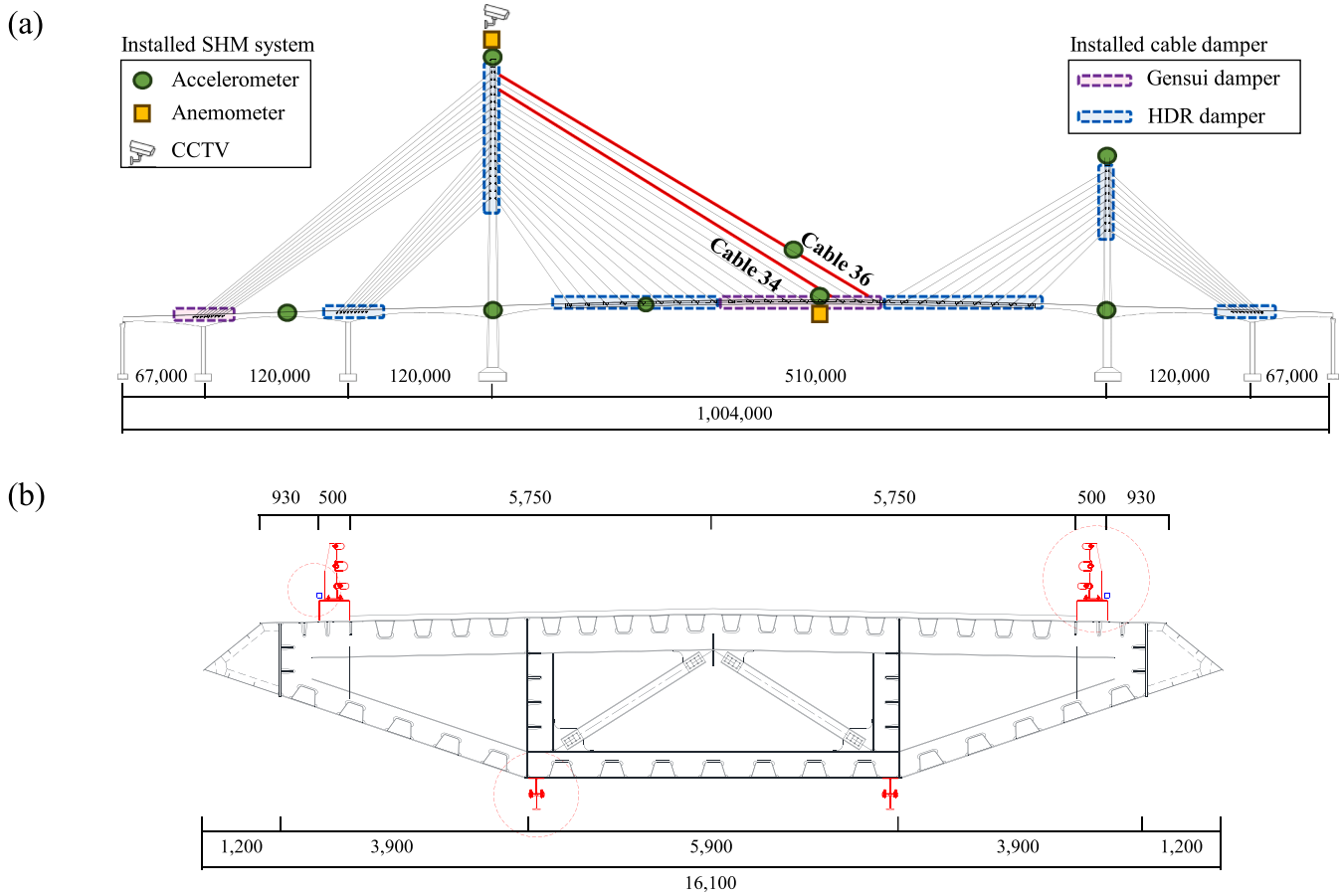


Fig. 1. The investigated cable-stayed bridge (Unit: mm); (a) Longitudinal arrangement; (b) Deck cross-section.



Fig. 2. CCTV images.

estimates of the bridge. The frequencies showed a daily trend averaging 0.316 and 0.435 Hz that aligned closely with the eigenfrequencies calculated from FE models of 0.313 and 0.444 Hz. While damping estimates exhibited significant scattering due to the nature of OMA-based damping estimates [17], the coefficients of variance (COV) for the two modes were within acceptable values at 27.95% and 19.24%, respectively. The averaged damping estimates were 0.343% for the V1 mode, which was lower than the design value of 0.4% [19], and 1.179% for the V2 mode, which surpassed the design value. These results aligned with

the observation that only the VIV in the V1 mode was observed on the bridge deck.

2.3. Observed VIV of the bridge deck and its suppression

Kim et al. [16] reported that the V1 mode VIV of the bridge deck occurred at the investigated bridge soon after its opening. Fig. 4 shows the vibrational characteristics and wind properties during a representative example of this VIV. The measured acceleration (Fig. 4(a) and

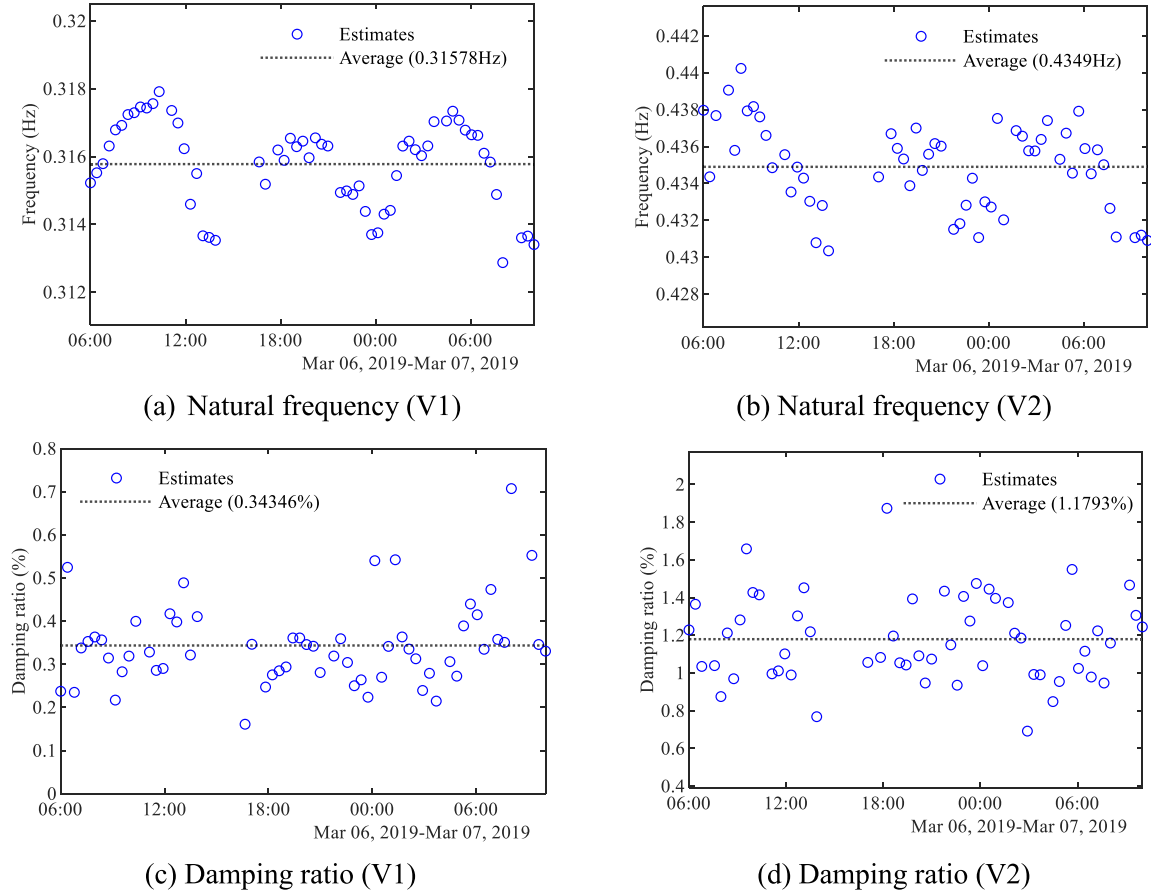


Fig. 3. Estimated modal parameters of the bridge.

Fig. 4(b)) exhibited significant vibrational amplitudes and clear harmonic motion. The maximum amplitude, approximately 1.13 m/s^2 (equivalent to 29.9 cm obtained through the displacement reconstruction algorithm), exceeded the serviceability limitation of 0.5 m/s^2 recommended in the Korean Design guidelines [19]. Fig. 4(c) further validates the observation of the narrow-banded characteristics of this vibration, where the predominant vibrational energy of the bridge deck is concentrated within the bandwidth of the V1 mode. Notably, the initiation of vibration coincided with a rapid decrease in wind speed after 2000 s, falling below 8 m/s (Fig. 4(d)). Such observations confirm the typical VIV characteristic, where harmonic motion occurs upon reaching the lock-in wind speed.

It was noted that the as-built bridge incorporated slightly different details in the inspection rails, both in terms of their geometrical shape and installation position. Additionally, disparities were observed between the as-built and designed shapes in the bridge railing and the attached small ducts, as indicated by the red circles in Fig. 1(b). The impact that such cross-sectional variation exerts on VIV is discussed via the results of wind tunnel tests in Section 3.2. The bridge owner and the contractor decided to install a MTMD inside the deck to suppress the VIV at the V1 mode of the bridge.

2.4. Observed stay cable vibrations afterwards

As previously mentioned in the Introduction, two types of cable vibrations continued even after the suppression of VIV in the bridge deck. One was the high-frequency vibrations in several cables, and the other was the low-frequency vibrations, particularly for Cable 34.

Fig. 5 presents representative examples of monitoring data for the first type of cable vibration at the longest stay cable (Cable 36). The

severe vibrations displayed maximum accelerations ranging from $30\text{--}50 \text{ m/s}^2$ at an average wind speed of approximately 10 m/s . It is essential to note that the accelerometer that was installed is rated with a measurement range of $\pm 50 \text{ m/s}^2$, and signal saturation occurred due to the cable's actual vibration exceeding the sensor's measurement boundaries. Note that, as shown in Fig. 1(a), the longest cables in each bundle group (e.g., Cable 36) were equipped with an accelerometer. Despite the flattened range in the frequency components resulting from this signal saturation, Fig. 5(c) and (d) clearly demonstrate a concentrated energy distribution of cable vibrations that averaged $22\text{--}26 \text{ Hz}$. This observation solidifies the categorization of these vibrations as traditional VIVs, transitioning between neighboring cable modes as the wind speed varied.

The excessive VIV in this specific cable could be attributed to a relatively low damping ratio specific to this mode. Kim et al. [16] reported that the estimated modal damping ratio within the frequency range of $20\text{--}25 \text{ Hz}$ was remarkably low, approximately 0.05% , and represented one of the lowest values reported for various stay cables [20]. Fig. 6 compares the nondimensional sag parameter (λ^2) that represents the impact of tension on the dynamic response of cables, with that of other cable-supported bridges of similar configuration. The sag parameter is defined by Eq. (1) [14].

$$\lambda^2 = \left(\frac{\rho g L \cos \theta}{T_0} \right) \frac{EAL}{T_0 L_e} \quad (1)$$

In Eq. (1), ρ , g , L , L_e , A , T_0 and θ are the cable mass per unit length, the gravitational acceleration, cable length, effective length of cable, cable section area, cable tension and inclination angle, respectively. Since the Seohae Bridge and Phu My Bridge do not have cables longer than 250 m , quadratic extrapolation was applied to predict the sag

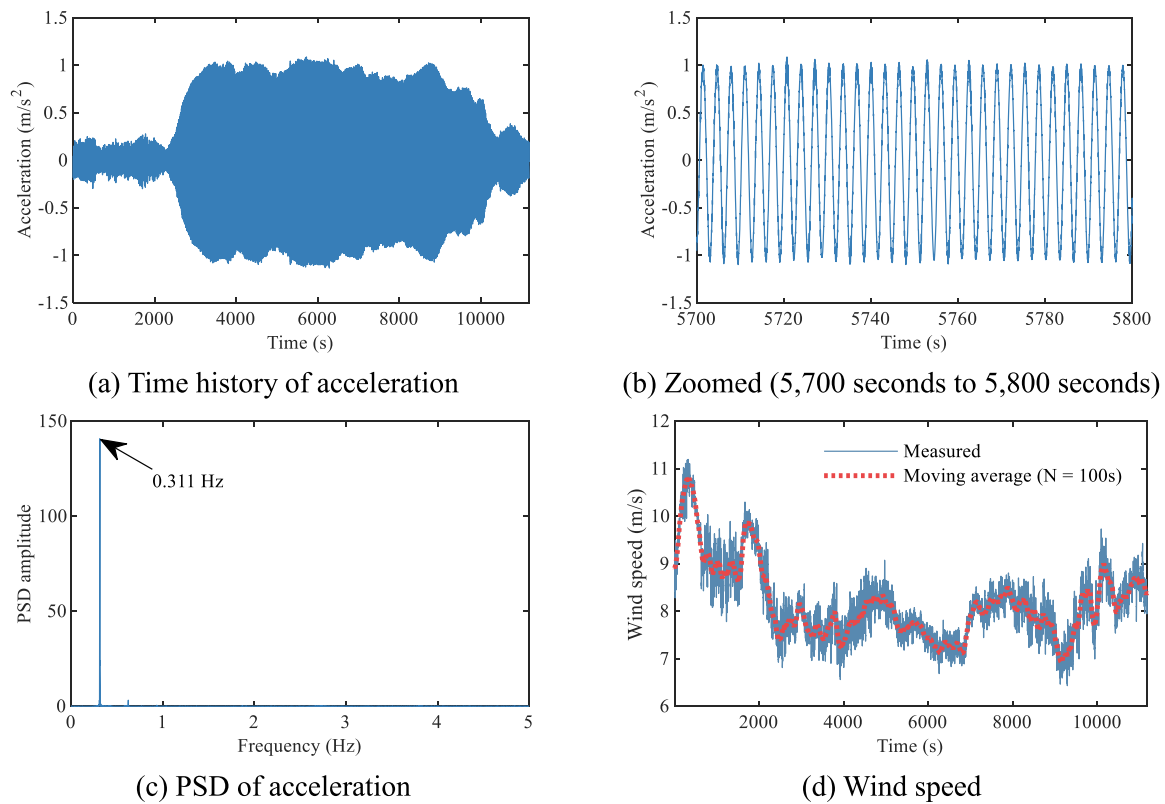


Fig. 4. The vibrational characteristics and wind properties under VIV occurrences on the bridge deck.

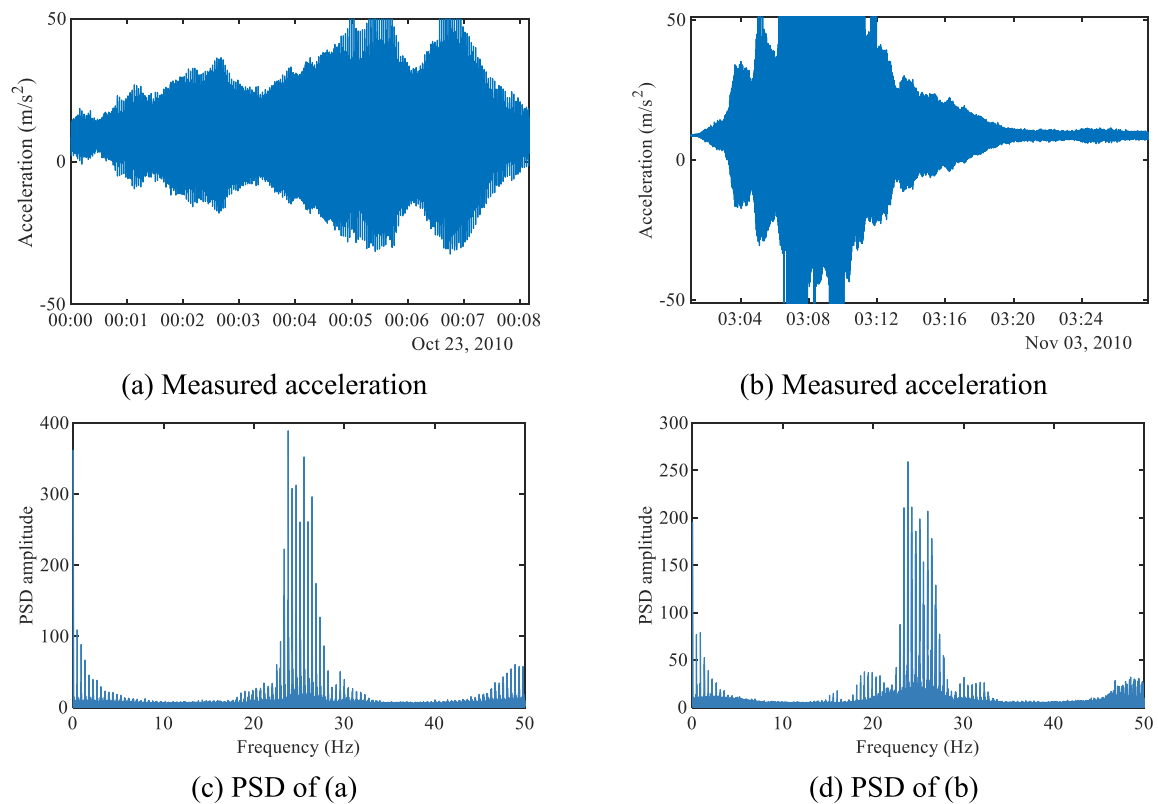


Fig. 5. The vibrational characteristics of high-mode VIVs for the longest cable.

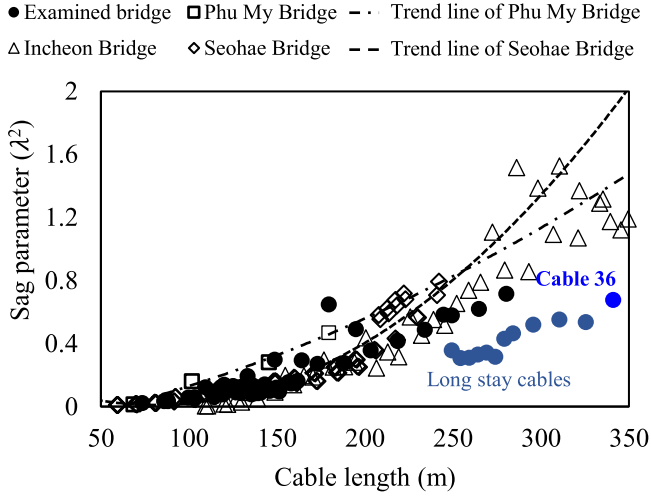


Fig. 6. Sag parameter according to the cable length
Retrieved from Kim et al. [16].

parameters for comparison purposes. Each extrapolation result is represented as a dashed line in Fig. 6. As shown in Fig. 6, the sag parameter of the investigated bridge demonstrated a significantly lower magnitude compared with that of other bridges with long cables that had not experienced hazardous cable VIVs. This suggests that the stay cables of the investigated bridge are under a relatively higher stress level in the dead load state. This condition may increase the vulnerability to VIVs compared to similar bridges. Further details can be found in the reference [16].

In addition to the analysis of sag parameter, the natural frequencies of the cables were also calculated, considering both the sag effect and cable bending stiffness. The natural frequency of cable, as proposed by Morse and Ingard [26], was calculated using Eq. (2), which accounts for cable bending stiffness:

$$f_{n_{bend}} = nf_0 \left[1 + 2\sqrt{\frac{EI}{T_0 L^2}} + \left(4 + \frac{n^2 \pi^2}{2} \right) \frac{EI}{T_0 L^2} \right] \quad (2)$$

where f_0 is the natural frequency calculated using taut string theory, EI is the bending stiffness of the cable, and $f_{n_{bend}}$ denotes the natural frequency of mode n considering the effect of cable bending stiffness.

Ahn and Lee [1] proposed the natural frequency equation for cables, which considers the effect of cable sag, as shown in Eq. (3).

$$f_{n_{sag}} = nf_0 \sqrt{1 + \frac{2\lambda^2}{(n\pi)^4} [1 - (-1)^n]^2} \quad (3)$$

where $f_{n_{sag}}$ denotes the natural frequency of mode n considering sag effect. By expanding the square root in Eq. (3) using a Taylor series expansion and neglecting higher-order terms, the natural frequency considering both sag effect and cable bending stiffness, $f_{n_{total}}$, can be estimated as follows:

$$f_{n_{total}} \approx nf_0 \left[1 + \frac{\lambda^2}{(n\pi)^4} [1 - (-1)^n]^2 + 2\sqrt{\frac{EI}{T_0 L^2}} + \left(4 + \frac{n^2 \pi^2}{2} \right) \frac{EI}{T_0 L^2} \right] \quad (4)$$

The parameters for Cables 34 and 36 are as follows: for Cable 34, $T_0 = 2110$ kN, $E = 200$ GPa, $I = 3.40 \times 10^{-6}$ m⁴, $L = 310.4$ m, $\lambda^2 = 0.551$; for Cable 36, $T_0 = 1958$ kN, $E = 200$ GPa, $I = 2.74 \times 10^{-6}$ m⁴, $L = 341.0$ m, $\lambda^2 = 0.676$.

Tables 1 and 2 present the calculated natural frequencies for several modes of Cables 34 and 36, respectively. Only for the first modes of both cables are affected by the consideration of the sag effect, while the taut string calculation was sufficiently accurate for the rest of the modes.

To examine the suppression of the high-mode VIV of cables, a series of Stock Bridge (SB) dampers were installed along the two cables (Cables 34 and 36), as shown in Fig. 7. The SB dampers were installed only on one side of the cable with respect to the cross-section of the deck, specifically cables 34 L and 36 L. This was done to compare with the motion of the bare cables, 34 R and 36 R. Fig. 8 depicts the velocity-amplitude curve of both sides of the cables during a typhoon. As shown in the figures, the SB dampers were effective in mitigating the high-mode VIV of the problematic cables, with the maximum acceleration during the event dropping to below 3.2 m/s².

During a periodic safety inspection, low-frequency cable vibration was occasionally observed for Cable 34. Directly measured acceleration data from Cable 34 were unavailable during the vibration event due to the absence of installed sensors. However, the analysis of low-quality CCTV images recorded from the top of the pylon visually confirmed 12 instances of significant cable vibrations. These vibrations were observed during the following time intervals: May 18, 2020 from 15:20 to 15:35; June 10, 2020, from 11:40 to 11:50, 12:05 to 12:10, and 15:14 to 15:20; July 13, from 01:46 to 02:10, and 05:28 to 05:37; July 19, 2020, from 13:45 to 13:55, and 14:20 to 14:30; August 3, 2020, from 10:34 to 10:40; August 4, 2020, from 14:00 to 14:30; and, August 5, 2020, from 11:24 to 11:32, and 15:23 to 15:30. To estimate the approximate vibration amplitude, we analyzed the pixel movement of the cable in the CCTV footage. One such instance revealed a vibrational amplitude of approximately 10 cm. Detailed information is provided in Section 3.4.

3. Cause investigation of low-frequency vibration

3.1. Monitored deck vibration during low-frequency cable vibration events

As previously mentioned, no accelerometer was installed on Cable 34 during the occurrence of low-frequency vibrations. Instead, to evaluate the low-frequency cable vibrations, the girder accelerations were measured at the center of the main span, where Cable 34 was anchored.

Representative monitoring data for the accelerometer and wind speeds from the center of the main span are presented in Fig. 9. Note that the acceleration data were plotted after passing through a 10 Hz low-pass filter to highlight the low-frequency vibration component, because the maximum modal frequency of the bridge deck was lower than 3 Hz. During the intervals highlighted in red in Fig. 9(a), harmonic motion with a maximum acceleration of approximately 6 cm/s² occurred, which was followed by a gradual decrease in vibration during

Table 1
Calculated natural frequencies for Cables 34.

Mode #	f_0 (Hz)	$f_{n_{sag}}$ (Hz)	$f_{n_{bend}}$ (Hz)	$f_{n_{total}}$ (Hz)	$\frac{f_0}{f_{n_{total}}}$	$\frac{f_{n_{sag}}}{f_{n_{total}}}$	$\frac{f_{n_{bend}}}{f_{n_{total}}}$
Mode 1	0.4223	0.4318	0.4238	0.4334	0.9743	0.9964	0.9779
Mode 2	0.8445	0.8445	0.8477	0.8477	0.9963	0.9963	1.0000
Mode 5	2.1113	2.1114	2.1199	2.1200	0.9959	0.9959	1.0000
Mode 11	4.6449	4.6449	4.6712	4.6712	0.9944	0.9944	1.0000

Table 2
Calculated natural frequencies for Cables 36.

Mode #	f_0 (Hz)	$f_{n_{\text{avg}}}$ (Hz)	$f_{n_{\text{bend}}}$ (Hz)	$f_{n_{\text{total}}}$ (Hz)	$\frac{f_0}{f_{n_{\text{total}}}}$	$\frac{f_{n_{\text{avg}}}}{f_{n_{\text{total}}}}$	$\frac{f_{n_{\text{bend}}}}{f_{n_{\text{total}}}}$
Mode 1	0.3849	0.3956	0.3861	0.3968	0.9701	0.9970	0.9731
Mode 2	0.7699	0.7699	0.7723	0.7723	0.9969	0.9969	1.0000
Mode 5	1.9247	1.9248	1.9313	1.9313	0.9966	0.9966	1.0000
Mode 11	4.2343	4.2343	4.2536	4.2536	0.9955	0.9955	1.0000



Fig. 7. Field installation of SB dampers.

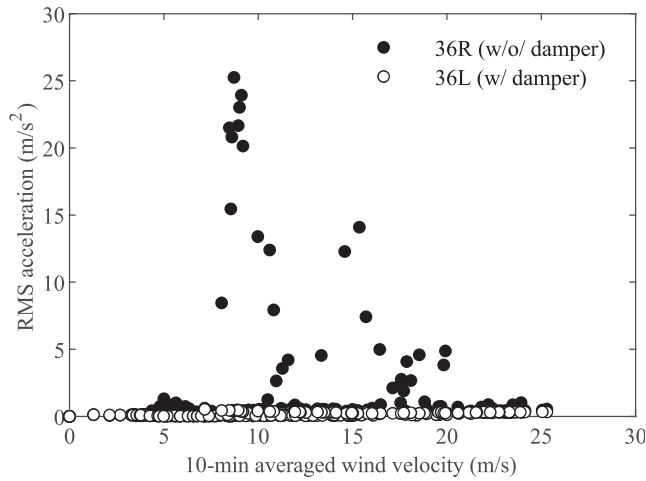


Fig. 8. V-A curve of Cable 36 according to the installation of SB dampers.

the intervals highlighted in blue.

Fig. 10 shows the power spectral densities (PSD) for the corresponding accelerations for each of the colored intervals in Fig. 9. As depicted in Fig. 10(a), the dominant vibration frequency during the red intervals is 0.433 Hz, corresponding to the V2 mode of the bridge. Following the disappearance of this narrow-banded response for the blue interval, Fig. 10(b) illustrates a broader-banded response, encompassing the first three vertical natural frequencies (0.310, 0.433, 0.583 Hz) of the bridge. While the first-frequency component of 0.310 Hz maintains a consistent level compared to Fig. 10(a), it now becomes dominant due to a significant decrease in the second-frequency component of 0.433 Hz. This broad-banded response, characterized by multiple mode components, typifies the motion behavior of a bridge under random vehicle operations. Examination of the wind speed (Fig. 9 (b)) during the occurrence of harmonic motion showed that a wind speed of approximately 10.5 m/s was maintained. This indicates that the low-frequency cable vibration could be related to the V2 mode vibration of the bridge deck.

Fig. 11 features a scatter diagram obtained from the monitoring data that shows the PSD amplitude of acceleration for the V2 mode of the

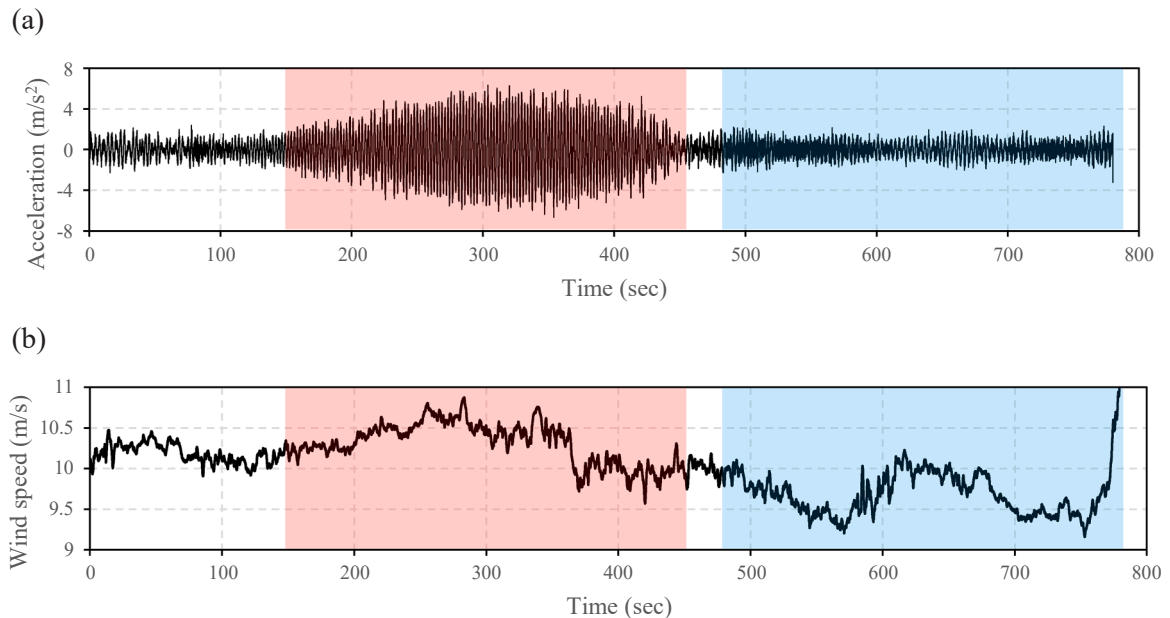


Fig. 9. Time series: (a) vertical acceleration; (b) wind speed at the center of the main span (July 19, 2020, from 13:46 to 13:59).

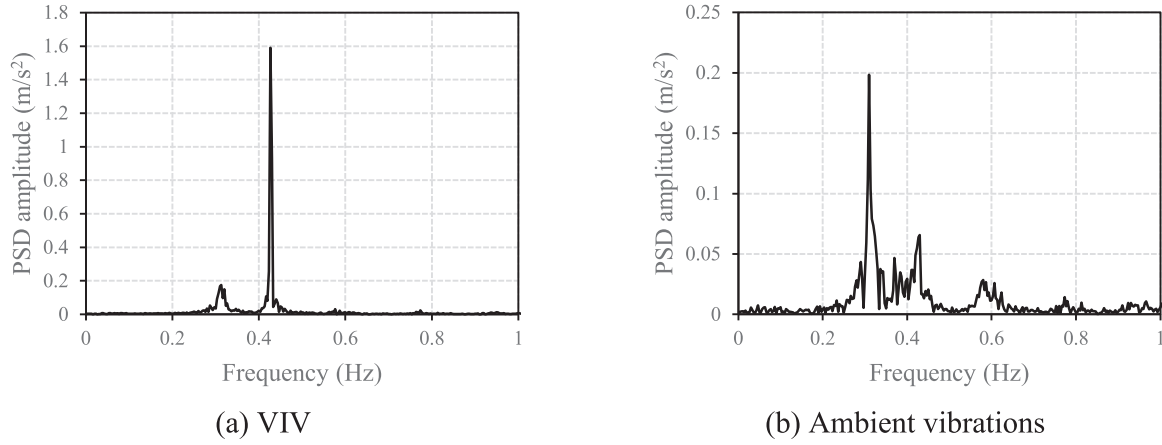


Fig. 10. PSD of measured acceleration at the center of the main span indicated in Fig. 9.

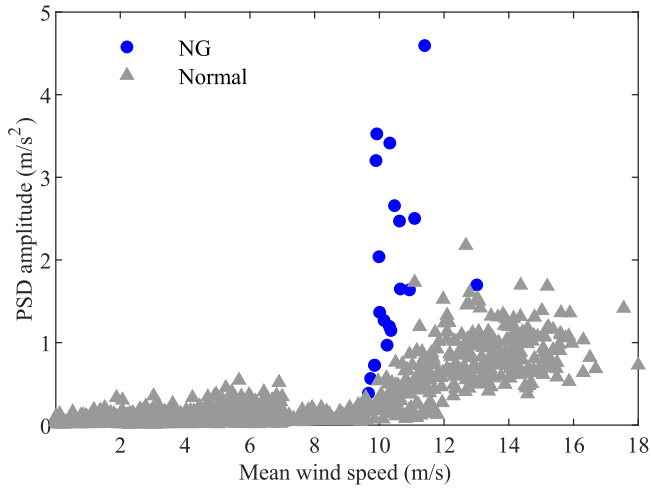


Fig. 11. PSD amplitude of bridge deck acceleration for the V2 mode according to the mean wind speed.

bridge deck along with the mean wind speed. The points labeled “NG” represent datasets associated with the 11 instances of low-frequency cable vibration manually identified through the CCTV inspection described in Section 2.4, while the points labeled “Normal” represent other instances. The observation that 11 NG cases deviate from the general increasing trend of the bridge deck acceleration, specifically of the V2 mode, with the increase of wind speed suggests a correlation between these NG cases of cable vibration and the exceptional deck motions of the V2 mode within the wind speed range of 10–12 m/s.

3.2. Aerodynamic investigation of the V2 mode vibration in the deck

This section describes a series of wind tunnel tests using the built-in sectional shape and modal properties of the bridge deck to assess the potential and reasons for VIV, which were not anticipated during the initial wind-resistance design phase. To this end, these experiments involved a comparison between the original design and as-built details,

Table 3
Parameters for wind tunnel test.

	Prototype	Target	Model
Width (B)	16.15 m	0.46 m	0.46 m
Mass	10.07 ton/m	8.22 kg/m	8.16 kg/m
Natural frequency	0.32 Hz	2.31 Hz	2.31 Hz
Damping ratio (ξ)	0.35%	-	0.10–0.95%

with details of the conditions summarized in Table 3. In order to scrutinize the variations in VIV amplitude that concern damping, experiments were conducted under five distinct damping ratio (ξ) conditions ranging from 0.10 to 0.95%.

Fig. 12 illustrates the results of the wind tunnel experiment. Even with a damping ratio of 0.1%, the original design did not exhibit significant vibrations, and the as-built cross-section suppressed the vibration only when the high damping ratio reached 0.95%. This comparison between the as-built and the original design highlights that the altered shape of the inspection rail in the as-built structure induced VIV along the deck.

These wind tunnel test results were further utilized to explain the cause of the V2 mode vibration of the bridge deck at a wind speed of 10.5 m/s. To extend these experimental findings of the fundamental mode to that of the higher vertical modes, the wind velocity (U) was normalized by dividing it by the bridge width (B) and the vibration frequency (n), which yielded a dimensionless reduced wind velocity. The VIV of the as-built cross-section was observed for a reduced wind velocity range of 1.5 to 1.8 in the wind tunnel test. Based on the test results, the Strouhal number (S_t), as defined by Eq. (5) using the shedding frequency (f_s), was determined to be 0.67.

$$S_t = \frac{f_s B}{U} \quad (5)$$

By employing the Strouhal number and examining the reduced wind velocity range associated with VIV, the shedding frequency was plotted against wind velocity, as depicted in Fig. 13. The expected lock-in wind speed for V2 ranged from 10.5 m/s to 12.6 m/s, which aligned with the recorded wind speed range for the “NG” datasets that varied from 9.66 to 13.01 m/s, as shown in Fig. 11. This finding emphasizes a significant correlation between the occurrences of cable vibration and the corresponding deck motion of the investigated bridge, particularly under windy conditions.

Additional analyses were performed to affirm the correlation between the actual measured deck VIV amplitude and the outcomes derived from wind tunnel tests. The VIV amplitude is dependent on the Scruton number (S_c), which is proportional to structural damping and the ratio between the vibrating mass and the mass of the air displaced by the structure, as defined by Eq. (6).

$$S_c = \frac{2\pi\xi m_e}{\rho B^2} \quad (6)$$

In Eq. (6), m_e is the equivalent modal mass per unit length and ρ is the air density. Fig. 14 shows the peak amplitude of VIV regarding the Scruton number, which was extracted from the experimental results in Fig. 12. The Scruton number and the maximum vertical VIV amplitude ($h_{\max}$) of the experimental data were fitted using an exponential curve,

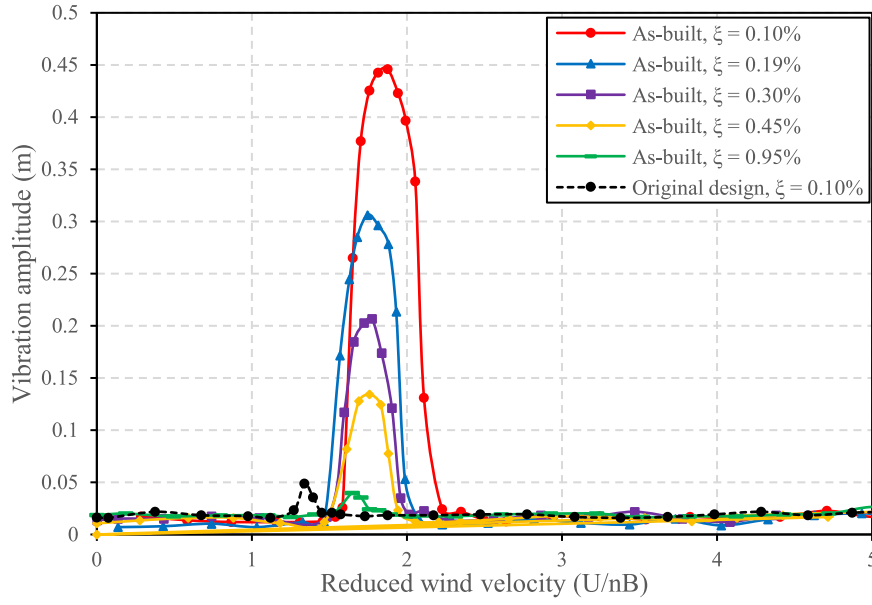


Fig. 12. Vibration-amplitude curve according to damping ratio.

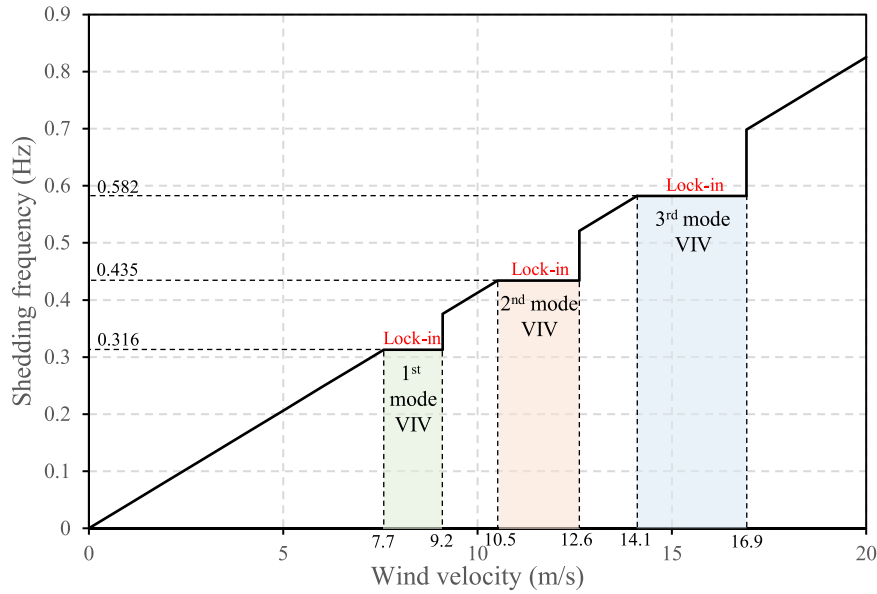


Fig. 13. Shedding frequency according to wind velocity.

as given by Eq. (7).

$$h_{\max} = 60.36e^{-1.785 S_c} \quad (7)$$

The 95% confidence limit curves are represented by the dashed lines as given in Eq. (8).

$$h_{\max} = (60.36 \pm 8.79)e^{(-1.7845 \pm 0.3895) S_c} \quad (8)$$

The Scruton number of the V2 mode was calculated as 2.44 with a damping ratio of 1.066% and an equivalent modal mass of 11.914 ton/m. The vibration amplitude of the deck in the V2 mode was estimated to be approximately 0.26 to 2.23 cm. To compare the experimental amplitudes with the acceleration data of the deck, the corresponding amplitude of displacement was estimated under the assumption of harmonic motion at a single frequency. The estimated vibration amplitude of the deck was 0.81 cm ($= (6\text{cm} \cdot \text{s}^{-2}) / (2\pi \times 0.434 \text{ Hz})^2$). This result is within the range of the predicted vibration amplitude of the

deck in the V2 mode.

3.3. Estimation of the cable amplitude due to the support excitations at the deck

The field measurement data and wind tunnel test results in Section 3.2 confirmed that the bridge deck was subjected to a small amount (maximum 0.8 cm in a single amplitude) of the V2 mode VIV during the low-frequency vibrations of Cable 34. Note that regardless of the insubstantial VIV amplitude of the deck, Cable 34 was subjected to a significant amount of vibration, which could be identified by the naked eye.

To assess the cause of such a unique phenomenon, a human-excitation test was conducted on both Cable 34 and its neighboring cables to extract their natural frequencies. For this purpose, supplementary accelerometers were deployed on the cables, and a vision-based measurement system utilizing a cellular phone camera was employed as

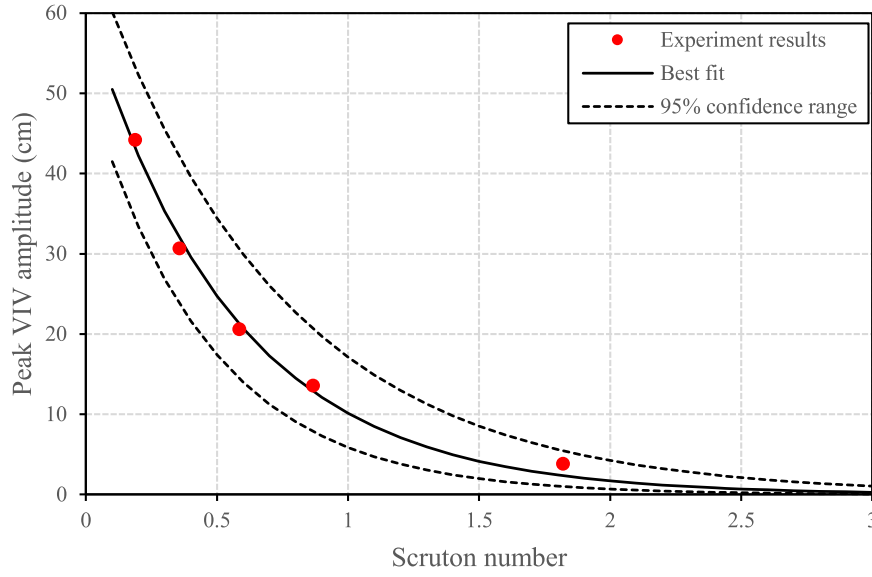
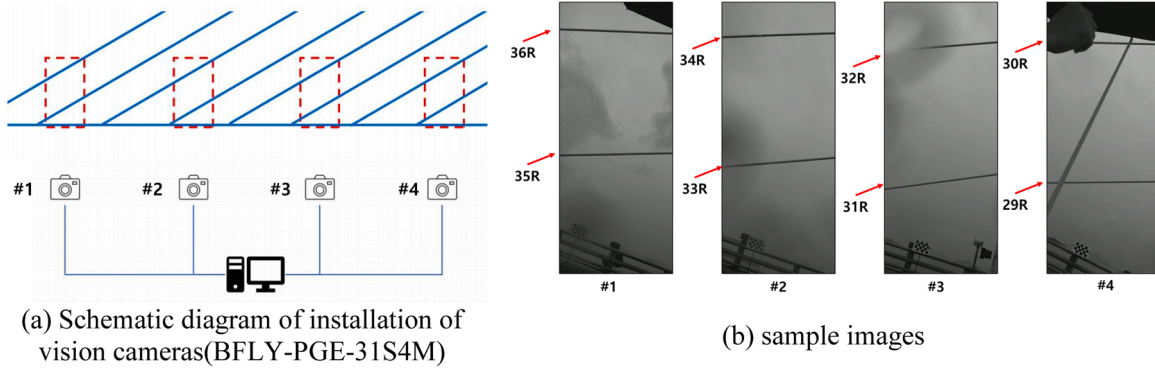
Fig. 14. VIV peak amplitude vs. S_c .

Fig. 15. Vision-based cable vibration measurement system.

well. The vision-based measurement system comprised four machine vision cameras (BFLY-PGE-31S4M) with resolutions of 2048×1536 and frame rates of 35 frames per second. The installation of the vision-based measurement system near the cable and the samples of recorded images are illustrated in Fig. 15. The tracking algorithm of Uni-KLT combined with Cable-ROI Net [13] was applied to those images for cable measurement. Additionally, multiple-view geometry was employed to minimize the distortions induced by the side-view camera angles [13, 21]. Last, the modal frequencies of Cable 34 were determined by identifying the peaks (peak-picking) in the PSD function of the cable vibration. As summarized in Table 4, the natural frequencies obtained from the vision-based system closely matched the accelerometer measurements. The calculated frequencies in Table 4 are confirmed to lie between the measured values obtained using the vision-based system and

the accelerometer, as shown in Table 4.

The structural damping ratio of the V1 mode of Cable 34 obtained from the long-term measurement data during the period from October 21, 2020, to November 11, 2020, was 0.40%. Aerodynamic damping acting on the cable during the vibration event was calculated using Eq. (9), as defined in the Korean Design guidelines [19].

$$\xi_{aero} = \frac{\rho D C_D}{8\pi f_c m_c} U \quad (9)$$

In Eq. (9), D is the diameter of the stay cable, C_D is the drag coefficient of the stay cable, f_c is the natural frequency of the stay cable, and m_c is the mass per unit length of the stay cable. Specific values used in the calculation were 0.7 for C_D , 0.095 m for D , and 30.7 kg/m for m_c . The aerodynamic damping at a wind speed of 10.5 m/s was calculated as 0.26%. The cable's total damping ratio (ξ_c) was estimated to be 0.66%.

Dynamic theory enables a prediction of the amplitude of cable vibration induced by deck motion. If the low-frequency cable vibration resulted from resonance phenomena, Korean Society of Civil Engineers [19] proposes an equation for calculating cable vibration amplitude (h_c), as given by Eq. (10).

$$h_c = \frac{2}{\pi} \frac{h_{\max}(f_d/f_c)^2}{\sqrt{[1 - (f_d/f_c)^2]^2 + [2\xi_c(f_d/f_c)]^2}} \quad (10)$$

In Eq. (10), f_d is the excitation frequency of the deck. Eq. (10)

Table 4

Estimated natural frequencies for Cable 34 using a vision-based measurement system.

Mode	Natural Frequencies (Hz)		
	Vision-based	Accelerometer	Difference
First mode	0.434	0.431	0.003
Second mode	0.838	0.869	0.031
Third mode	1.260	1.303	0.043
Fourth mode	1.676	1.741	0.065
Fifth mode	2.098	2.180	0.082

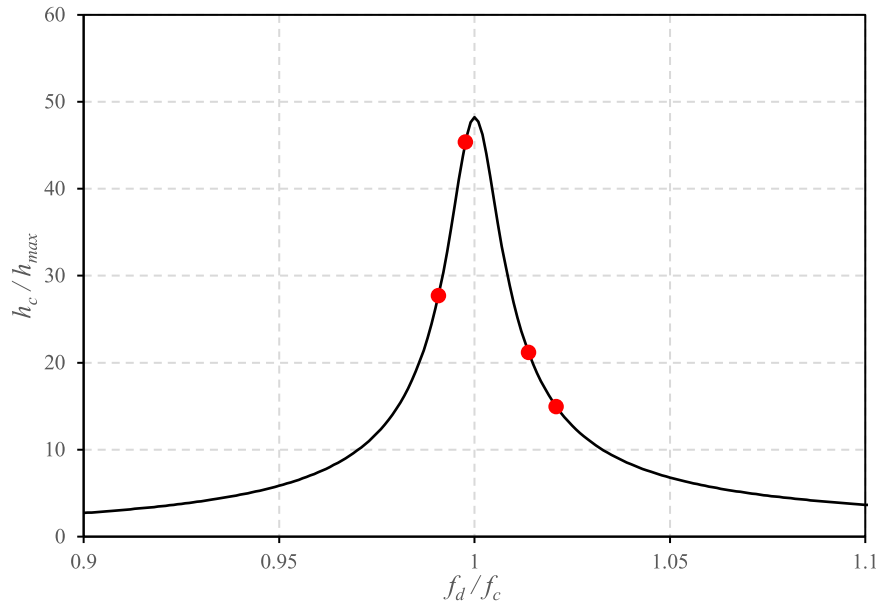


Fig. 16. Changes in cable vibration amplitude with respect to the frequency ratio.

indicates that the ratio f_d/f_c is crucial in determining the stay cable amplitude due to support excitation. The variation in cable vibration amplitude with respect to the frequency ratio is shown in Fig. 16. The red points correspond to values calculated using the first mode natural frequency of Cable 34 measured with accelerometers and video-based methods, as well as the estimated natural frequency of the V2 mode, as discussed in Section 2.2.

Based on the assumption that the vibration amplitude of the deck is 0.81 cm, the expected vibration amplitude of the cable ranges from 12.1 to 36.8 cm, which corresponds to a range of 89.0 to 269.6 cm²/s when converted to acceleration with the harmonic vibration assumption of the VIV. Since frequency of the V1 mode of Cable 34 (=0.431–0.434 Hz) was similar to that of the V2 mode of the bridge (=0.430–0.440 Hz), the vibration amplification of the cable could be amplified by a factor of 15- to 45-fold that of the deck.

The estimated amplitude of deck vibration, 0.81 cm, is below the allowed vibration amplitude specified in the Korean Design guidelines [19]. Calculations based on serviceability limitations indicate that the permissible amplitude is 6.7 cm ($= (0.5 \text{ m} \cdot \text{s}^{-2}) / (2\pi \times 0.434 \text{ Hz})^2$), assuming an acceleration limit of 0.5 m/s². However, even this small level of deck vibration can be significantly amplified in the cables, leading to large cable vibrations. Therefore, it is questionable whether this serviceability limit is appropriate for determining the admissible VIV amplitude. The authors believe that the serviceability limit should be applied to the bridge vibrations caused by vehicle load or buffeting-induced vibrations up to around 25 m/s. The potential VIV should be addressed through appropriate measures.

3.4. Field validation of the estimated support excitation

Finally, this section describes the field validation of the empirically estimated support excitations discussed in Section 3.3. As previously mentioned, no equipment or sensors were installed to measure the dynamic responses of Cable 34. Although a short-term monitoring campaign utilizing the vision-based system was conducted, no considerable V2 mode of deck vibration occurred during that period. In other words, there is no available data from simultaneous monitoring of neither the wind conditions, the V2 mode deck vibration, nor the dynamic responses of Cable 34.

To address these limitations, we quantified the cable pixel movement in sequential CCTV images instead of by direct measurement. Following

the approach suggested by Lee et al. [22], the process involves four steps: (1) determining the Region of Interest (RoI), (2) extracting the cable region (foreground) through an image binarization process, (3) removing the remaining outliers due to dynamic backgrounds and estimating the centerline of foreground pixels (inliers) using Random Sample Consensus (RANSAC), and (4) quantifying cable pixel movement by tracking the location of the y-intercept of the centerline. The schematic diagram of this algorithm is depicted in Fig. 17. Note that the RoI is not centered on the cable, and the CCTV angle may introduce out-of-plane motion, which cannot be adequately estimated in a two-dimensional, single-angle image. Therefore, the quantified pixel movement of the cable in this study is inherently underestimated.

Fig. 18 presents the time histories of the cable pixel movement, V2 mode displacement of the bridge deck, and wind speed. The pixel movement (y) was derived from sequential CCTV images recorded during the period of May 18, 2020 from 15:20 to 15:40 and was normalized by the pixel number corresponding to the cable diameter in the images (D). Note that camera calibration and precise vibration quantification are not feasible in this study due to the lack of information on camera positions or targets. Therefore, this study utilizes the normalized movement (y/D), which represents the relative motion of a cable pixel compared to the number of pixels in the cable diameter. The V2 mode displacement of the bridge deck was determined using FFIR-based displacement reconstruction of the bandpass-filtered acceleration (bandwidth = 0.35–0.5 Hz). The wind speed is depicted with moving averaged values to enhance the clarity of overall variation trends.

As the wind speed decreased below a lock-in wind speed of 12 m/s at around 15:14 on May 18, 2020, there was an onset of V2 mode vibration on the deck. At around 15:20 on May 18, 2020, when the wind speed was approximately 10 m/s, the V2 mode vibration on the deck exhibited a considerable vibration amplitude, and the pixel movement of Cable 34 gradually increased as well. When the vibration amplitude of the bridge deck peaked at 15:26 on May 18, 2020, the cable amplitude also increased to approximately 1.0 times the cable diameter (referred to as 1.0D). With a cable diameter of 9.5 cm, the normalized pixel movement roughly measured about 10.001 cm (1.05D). The ratio between the cable vibration and bridge deck of 0.501 cm was subsequently 19.97. Considering the potential underestimation of cable vibration, this ratio fell to within the range derived in Section 3.3, approximately 15- to 45-fold. These field observations not only confirm the earlier empirical

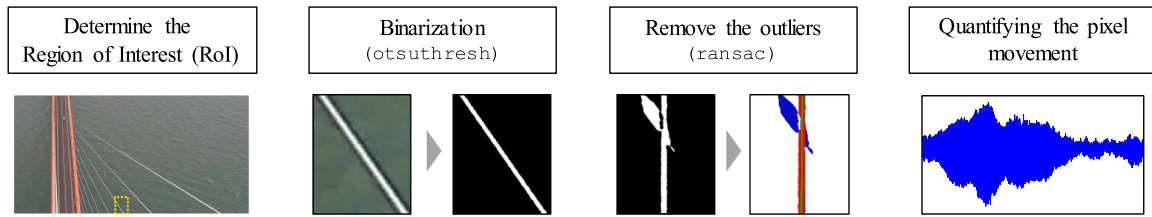


Fig. 17. Schematic diagram for a CCTV-based quantification of cable movement.

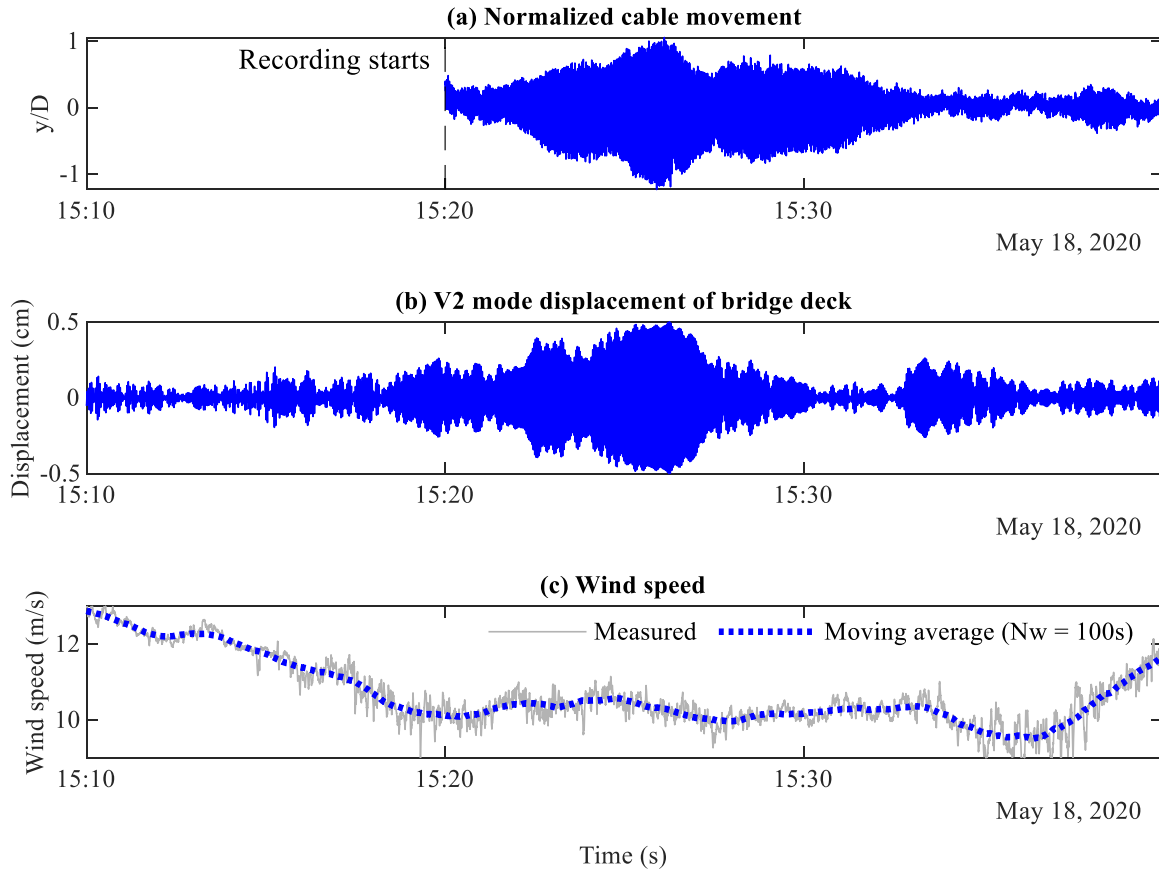


Fig. 18. Time histories of monitoring datasets representing low-frequency cable vibration.

analysis results but also demonstrate that even minor vibrations on the bridge deck can induce detrimental cable vibrations exceeding 1D.

4. Conclusions

This study investigated extraordinary amounts of low-frequency vibration observed in a cable-stayed bridge. Based on the FE analysis, OMA, CCTV video analysis, and the vision-based vibration monitoring, the source of the cable vibration was the small amount of the deck VIV in a V2 mode at a lock-in wind speed of 10.5 m/s at the deck level. In fact, the VIV problem of the deck was initiated from the improper installation of inspection rails under the steel box deck on the main span of the bridge. Even though the main VIV was mitigated by the MTMD installed after opening of the bridge, the V2 mode VIV eventually excited Cable 34, although the amplitude was less than 1 cm. This is a rare report of an observation of cable vibration due to support excitation. The very close frequencies between the cable V1 mode and that of the bridge V2 mode also contributed to the observation of low-frequency cable vibration.

Based on the shared data and cause investigation, the following managerial insights are drawn.

- 1) The potential coincidence check between the natural frequencies between the deck mode and cable modes of all stays is important. This is a conventional item in a design checklist of potential cable vibration. The present investigation, however, should serve as a reminder of the importance of the resonance phenomenon.
- 2) From the viewpoint of the vibrational serviceability of long-span cable-stayed bridges, the deck vibration level is limited to a certain acceptable level in maximum or root mean square acceleration. The Korean Society of Civil Engineers [19] recommends a value of 0.5 m/s^2 for the maximum acceleration of the deck. This criterion is conventionally used as an acceptable level of VIV of the deck during wind tunnel test. However, when there is a high potential for frequency coincidence between the deck and the cable mode in a cable-stayed bridge, the suppression of even a small level of VIV based on aerodynamical shape investigation with additional fairing or flap devices is recommended.

CRediT authorship contribution statement

Sangsub Ahn: Methodology, Investigation, Formal analysis.

Sunjoong Kim: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Hyungchul Yoon:** Writing – original draft, Investigation, Formal analysis. **Jaeyeong Lim:** Investigation, Formal analysis, Data curation. **Ho-Kyung Kim:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Youchan Hwang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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